Ariana Ariaei

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Education

2022 - present Bachelor of Science in Computer Engineering at **Babol Noshirvani University of Technology**
(Ranked $501 - 600^{th}$ in World University Rankings 2025)
GPA: 3.6/4.0 over last 2 years

Selected Courses

Principles of Compiler Design Score: 19.6/20
Instructed by: Dr. A. Gholami Rudi

Fundamentals and Applications of Artificial Intelligence Score: 17.1/20
Instructed by: Dr. F. Zamani

Formal Languages & Automata Theory Score: 18.5/20
Instructed by: Dr. M. Hassanpour

Computer Architecture Score: 18/20
Instructed by: Dr. M. Valinataj

Applied Linear Algebra Score: 19.25/20
Instructed by: Dr. S. Jamshidpour

Software Systems Analysis and Design Score: 19.6/20
Instructed by: Dr. M. Emadi

Awards and Honors

- **Top 6% in the Iranian University Entrance Exam**: Achieved a ranking among the top 6% of participants in the highly competitive Iranian university entrance exam, which is considered one of the most challenging academic assessments in the country. This recognition allowed for admission to a prestigious university.
- **University Scholarship (Tuition Waiver)**: Awarded a full scholarship for academic excellence, covering all tuition fees for my Bachelor's degree program at Babol Noshirvani University of Technology. The scholarship was granted based on my outstanding performance in the entrance exam and my academic achievements.
- **Ranked Top 6% in the 6^{th} AlgoNIT National Programming Contest**: Placed among the top 6% of 167 participants in the 6^{th} AlgoNIT National Programming Contest, demonstrating strong problem-solving skills and competitive programming expertise.
- **Ranked 1^{st} at high school in the Iranian University Entrance Exam**: Among fellow students specializing in Mathematics and Physics.

Research Interests

- Machine Learning
- Deep Learning
- Computer Vision
- Large Language Models (LLMs)
- Artificial Intelligence
- Neural Networks
- AI in Healthcare
- Deep Learning for Medical Imaging

Publications

The Role of Machine Learning in Diagnosis of Acute Occlusion Myocardial Infarction in Presence of Left Bundle Branch Block (LBBB) *In Progress*

Detecting myocardial infarction in the presence of LBBB is a challenging task even for experienced clinicians. This research aims to integrate AI and machine learning algorithms to automate the diagnostic process and potentially discover novel criteria for accurate detection.

AI-Driven Multimodal Analysis of Stress Responses in Virtual Reality Simulations *In Progress*

This project involves creating a stressful simulation using the Unity game engine and virtual reality (VR). Participants' multimodal data — including physiological signals (EEG, ECG, EMG, etc.), facial expressions, postures, and decision-making patterns — are monitored. The collected data will be analyzed using AI and large language models (LLMs) to uncover correlations between stress, physiological responses, and behavioral outcomes.

Academic Projects

Scam Sleuth: AI-Powered Scam Reporting and Detection Platform *2024–2025*

Designed and implemented a platform for detecting and reporting scam projects, featuring secure user authentication, an AI-powered website trustworthiness checker, and an admin dashboard for managing reports and publishing informative blog posts. Users can submit scam reports, analyze suspicious URLs, and engage through comments.

Tslator: A Compiler for TSLANG *2025*

Developed a compiler for the educational C-like language **TSLANG** under the supervision of Dr. Gholami Rudi. The project implemented lexical analysis, parsing, semantic analysis, and intermediate code generation using Python and PLY. Key features include support for basic data types, functions, arrays, conditionals, loops, and string handling, with an Abstract Syntax Tree, scoped symbol tables, type checking, and IR code generation.

PyThreadServe: Multi-Process HTTP/1.0 Server with Concurrency Control *2025*

Implemented a lightweight HTTP/1.0 server in Python with multi-process architecture, worker processes, and thread pooling. The server supports GET and POST requests, round-robin load balancing, and concurrency management with a cap on simultaneous POST requests. Additional features include thread-safe file operations, directory traversal protection, and comprehensive request logging.

Experiences

Research Experience

- **Research Assistant at Artificial Intelligence Research Lab, Babol Noshirvani University of Technology** *2025–present*
  Engaged in research projects focusing on software systems and applications of artificial intelligence.

Teaching Experience

- **Teaching Assistant for Principles of Compiler Design** *2025*
  Instructed by: Dr. A. Gholami Rudi

- **Teaching Assistant for Applied Linear Algebra** *2024*
  Instructed by: Dr. J. Kazemitabar
- **Teaching Assistant for Data Structures** *2024*
  Instructed by: Dr. H. Omranpour
- **Teaching Assistant for Introduction to Computer Science and Programming** *2023*
  Instructed by: Dr. H. Omranpour

Skills

Computer Skills

- **Programming Languages & Frameworks:** Python, C#, .NET, C++, VHDL, Matlab, Shell Script
- **Machine Learning Libraries:** Pytesseract, NumPy, Pandas, Scikit-learn, OpenCV, Matplotlib, LangChain
- **Technologies:** Unity Engine, Docker, Git, LaTeX

Language Skills

- **English:** Preparing for IELTS exam (to be taken on October 31st)

Hobbies and Interests

- Playing the piano
- Playing the guitar
- History
- Embroidery

References

Dr. Ali Gholami Rudi
Assistant Professor, Faculty of Electrical & Computer Engineering, Babol Noshirvani University of Technology
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Dr. Seyed Javad Kazemitabar
Associate Professor, Faculty of Electrical & Computer Engineering, Babol Noshirvani University of Technology
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